CPU Scheduling Simulator

Round Robin vs SJF (Preemptive) — Test Case Report

This document presents the four pre-loaded test scenarios used to validate the Round Robin (RR) and Shortest Job First Preemptive (SRTF) scheduling simulator. For each scenario the input workload, Gantt chart execution timelines, ready-queue snapshots, per-process metrics (WT, TAT, RT) and a side-by-side comparison summary are shown.

Scenario	Description	Quantum	Processes
A – basic mixed	Mixed burst times, staggered arrivals	3	4
B – short-job heavy	6 processes, mostly short bursts	2	6
C – fairness test	All equal burst times	4	4
D – long job	One dominant long job vs short arrivals	3	4

Scenario A – Basic Mixed

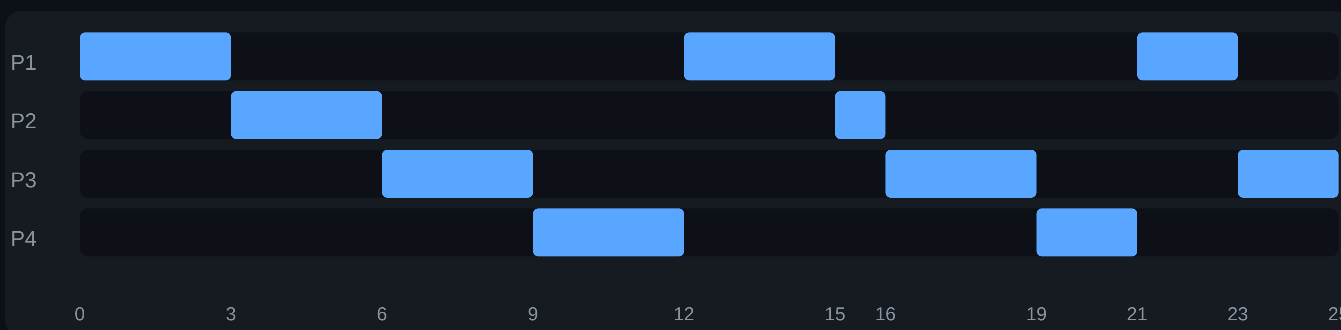
A general workload with four processes having different burst times and staggered arrival times. Tests the core scheduling behaviour of both algorithms under typical conditions.

INPUT

`Time Quantum (Round Robin) = 3 | Processes: 4`

PID	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	8
P4	3	5

GANTT CHART — ROUND ROBIN



READY QUEUE SNAPSHOTS — ROUND ROBIN

t0: P1	**t3: P2**	**t6: P3**	**t9: P4**	**t12: P1**
t15: P2	**t16: P3**	**t19: P4**	**t21: P1**	**t23: P3**

GANTT CHART — SJF PREEMPTIVE (SRTF)

METRICS

METRICS — ROUND ROBIN	METRICS — SJF (PREEMPTIVE)

PID	WT	TAT	RT
P1	15	23	0
P2	11	15	2
P3	15	23	4
P4	13	18	6
avg	**13.5**	**19.75**	**3.0**

PID	WT	TAT	RT
P1	9	17	0
P2	0	4	0
P3	15	23	15
P4	2	7	2
avg	**6.5**	**12.75**	**4.25**

COMPARISON SUMMARY

Metric	Round Robin	SJF Preemptive
avg waiting time	13.5	**6.5 ✓**
avg turnaround time	19.75	**12.75 ✓**
avg response time	**3.0 ✓**	4.25

Analysis: For this scenario, **SJF** achieves lower average waiting time (6.5 vs 13.5), **SJF** achieves lower average turnaround time (12.75 vs 19.75), and **RR** achieves lower average response time (3.0 vs 4.25).

Scenario B – Short-Job Heavy

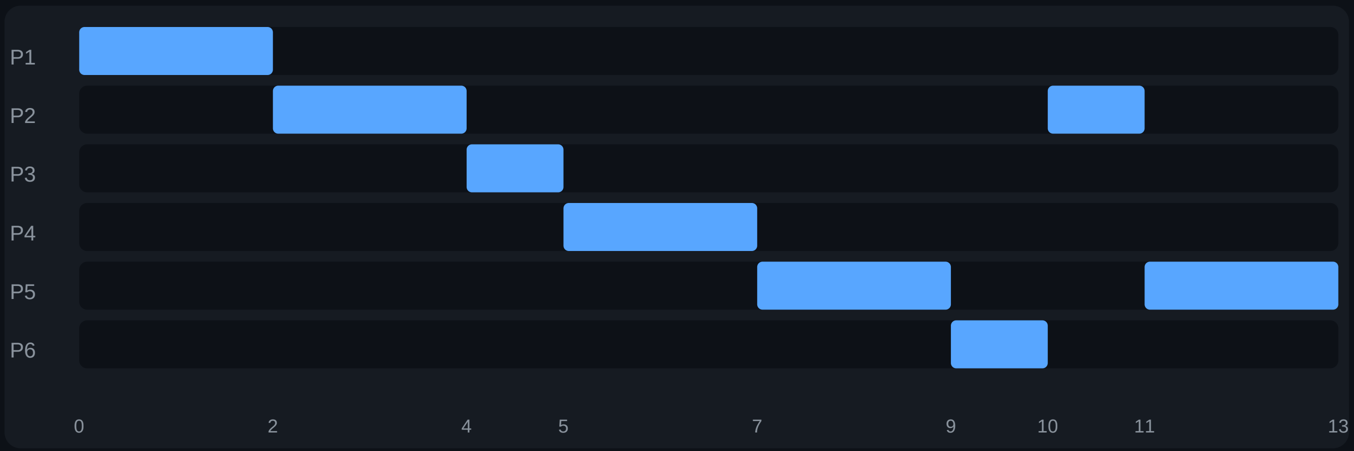
Six processes where most burst times are very short (1–4 units). Designed to reveal how SJF Preemptive takes advantage of short jobs compared to Round Robin's fixed-quantum rotation.

INPUT

`Time Quantum (Round Robin) = 2 | Processes: 6`

PID	Arrival Time	Burst Time
P1	0	2
P2	0	3
P3	1	1
P4	2	2
P5	3	4
P6	4	1

GANTT CHART — ROUND ROBIN



READY QUEUE SNAPSHOTS — ROUND ROBIN

t0: P1	**t2: P2**	**t4: P3**	**t5: P4**	**t7: P5**
t9: P6	**t10: P2**	**t11: P5**		

GANTT CHART — SJF PREEMPTIVE (SRTF)

METRICS

METRICS — ROUND ROBIN			
PID	**WT**	**TAT**	**RT**
P1	0	2	0
P2	8	11	2
P3	3	4	3
P4	3	5	3
P5	6	10	4
P6	5	6	5
avg	**4.17**	**6.33**	**2.83**

METRICS — SJF (PREEMPTIVE)			
PID	**WT**	**TAT**	**RT**
P1	0	2	0
P2	6	9	6
P3	1	2	1
P4	1	3	1
P5	6	10	6
P6	1	2	1
avg	**2.5**	**4.67**	**2.5**

COMPARISON SUMMARY

Metric	Round Robin	SJF Preemptive
avg waiting time	4.17	**2.5 ✓**
avg turnaround time	6.33	**4.67 ✓**
avg response time	2.83	**2.5 ✓**

Analysis: For this scenario, **SJF** achieves lower average waiting time (2.5 vs 4.17), **SJF** achieves lower average turnaround time (4.67 vs 6.33), and **SJF** achieves lower average response time (2.5 vs 2.83).

Scenario C – Fairness Test

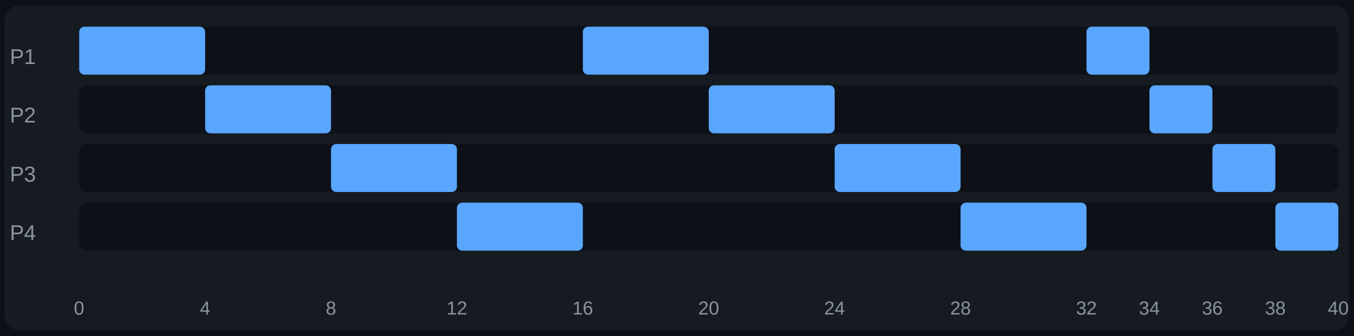
All four processes have identical burst times (10 units each) with consecutive arrivals. Exposes fairness behaviour: RR distributes CPU evenly, while SRTF prioritises whichever arrived first.

INPUT

`Time Quantum (Round Robin) = 4 | Processes: 4`

PID	Arrival Time	Burst Time
P1	0	10
P2	1	10
P3	2	10
P4	3	10

GANTT CHART — ROUND ROBIN



READY QUEUE SNAPSHOTS — ROUND ROBIN

`t0: P1`	`t4: P2`	`t8: P3`	`t12: P4`	`t16: P1`
`t20: P2`	`t24: P3`	`t28: P4`	`t32: P1`	`t34: P2`

GANTT CHART — SJF PREEMPTIVE (SRTF)

METRICS

METRICS — ROUND ROBIN	**METRICS — SJF (PREEMPTIVE)**

PID	WT	TAT	RT
P1	24	34	0
P2	25	35	3
P3	26	36	6
P4	27	37	9
avg	**25.5**	**35.5**	**4.5**

PID	WT	TAT	RT
P1	0	10	0
P2	9	19	9
P3	18	28	18
P4	27	37	27
avg	**13.5**	**23.5**	**13.5**

COMPARISON SUMMARY

Metric	Round Robin	SJF Preemptive
avg waiting time	25.5	**13.5 ✓**
avg turnaround time	35.5	**23.5 ✓**
avg response time	**4.5 ✓**	13.5

Analysis: For this scenario, **SJF** achieves lower average waiting time (13.5 vs 25.5), **SJF** achieves lower average turnaround time (23.5 vs 35.5), and **RR** achieves lower average response time (4.5 vs 13.5).

Scenario D – Long Job

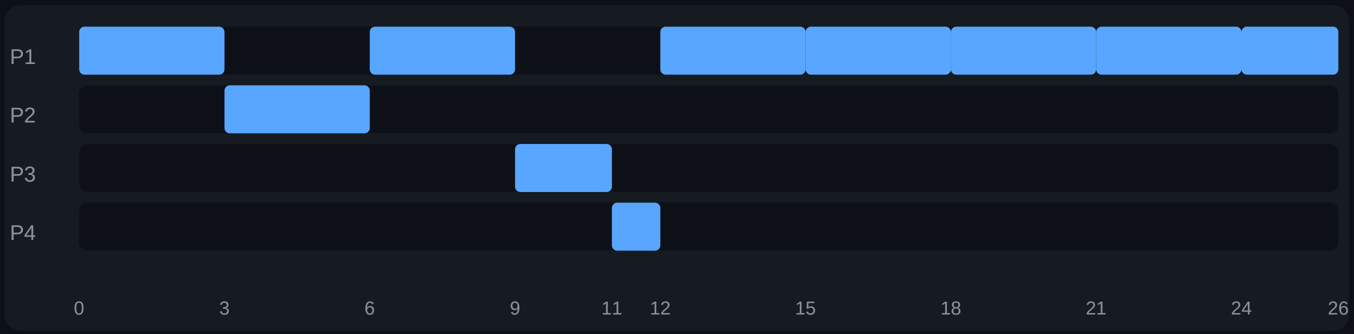
One very long process (P1, burst=20) arrives first, followed by three short processes. Demonstrates how SRTF preempts the long job as shorter processes arrive, while RR gives each process a fair time slice regardless of length.

INPUT

`Time Quantum (Round Robin) = 3 | Processes: 4`

PID	Arrival Time	Burst Time
P1	0	20
P2	2	3
P3	4	2
P4	6	1

GANTT CHART — ROUND ROBIN



READY QUEUE SNAPSHOTS — ROUND ROBIN

`t0: P1`	`t3: P2`	`t6: P1`	`t9: P3`	`t11: P4`
`t12: P1`	`t15: P1`	`t18: P1`	`t21: P1`	`t24: P1`

GANTT CHART — SJF PREEMPTIVE (SRTF)

METRICS

METRICS — ROUND ROBIN	METRICS — SJF (PREEMPTIVE)

PID	WT	TAT	RT
P1	6	26	0
P2	1	4	1
P3	5	7	5
P4	5	6	5
avg	**4.25**	**10.75**	**2.75**

PID	WT	TAT	RT
P1	6	26	0
P2	0	3	0
P3	1	3	1
P4	1	2	1
avg	**2.0**	**8.5**	**0.5**

COMPARISON SUMMARY

Metric	Round Robin	SJF Preemptive
avg waiting time	4.25	**2.0 ✓**
avg turnaround time	10.75	**8.5 ✓**
avg response time	2.75	**0.5 ✓**

Analysis: For this scenario, **SJF** achieves lower average waiting time (2.0 vs 4.25), **SJF** achieves lower average turnaround time (8.5 vs 10.75), and **SJF** achieves lower average response time (0.5 vs 2.75).